

## Chapter 12 - Vectors

1. Review of Composition and Resolution of Vectors
  - a. Scalars and Vectors
  - b. Notation and Representation
  - c. Vectors in ordered pair form
    - i. Position vector
    - ii. Displacement vector
  - d. Modulus of a vector
  - e. Different types of vectors
    - i. Unit vector
    - ii. Null/zero vector
    - iii. Negative vector
    - iv. Like vector
    - v. Unlike vector
    - vi. Equal vectors
  - f. Addition of Two Vectors
    - i. Triangle Law of Vector Addition
  - g. Difference of Two Vectors
  - h. Multiplication of a vector by a scalar
  - i. Laws of vector algebra
    - i. Commutative law
    - ii. Associative law
    - iii. Distributive law
  - j. The unit vectors  $\mathbf{i}$ ,  $\mathbf{j}$ , and  $\mathbf{k}$
2. Collinear and Non-collinear Vectors
  - a. Condition of Collinearity
3. Coplanar and Non-coplanar Vectors
  - a. Condition of Coplanarity
4. Linear Combination of Vectors
5. Linearly Dependent and Independent Vectors
  - a. Condition for Linear Dependency and Independency

## #Review of Composition and Resolution of Vectors

### Scalars and Vectors

|                |                                                                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scalars</b> | Scalars (or scalar quantity) are the physical quantities which have only magnitudes but no direction.<br>E.g., mass, density, time, speed, etc.          |
| <b>Vectors</b> | Vectors (or vector quantity) are the physical quantities which have both magnitude and direction.<br>E.g., velocity, acceleration, momentum, force, etc. |

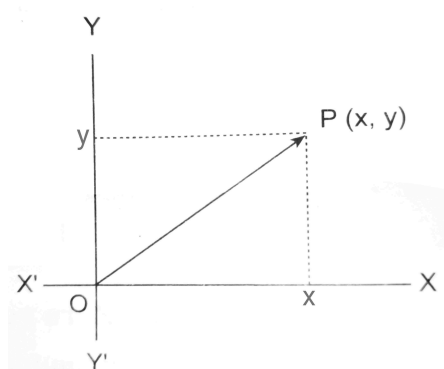
### Notation and Representation

Traditionally, vectors were represented by bold faced letters like **a**, **b**, **AB**, etc. But now, they are represented by a letter or combination of two letters with an arrow over it; such as  $\vec{A}$ ,  $\vec{B}$ ,  $\vec{AB}$ , etc.

But a letter or combination of two letters without an arrow means a scalar.

### Vectors in ordered pair form

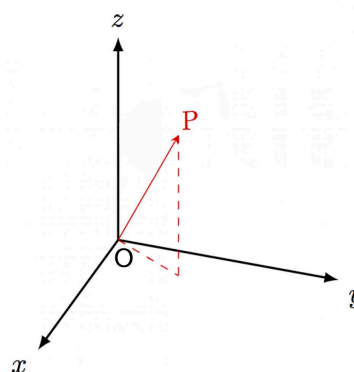
#### Position vector



Let  $P(x, y)$  be a point in the  $XY$  plane then

$$\vec{OP} = \begin{pmatrix} x \\ y \end{pmatrix} = (x, y)$$

This vector is known as the plane vector.



Similarly, If  $P(x, y, z)$  is a point in the space then

$$\vec{OP} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = (x, y, z)$$

This vector is known as the space vector.

#### Displacement vector

If  $A(x_1, y_1)$  and  $B(x_2, y_2)$  are two points in a plane then the displacement vector  $\vec{AB}$  is

$$\vec{AB} = \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix} = (x_2 - x_1, y_2 - y_1)$$

Similarly, If  $A(x_1, y_1, z_1)$  and  $B(x_2, y_2, z_2)$  are two points in the space then the displacement vector  $\vec{AB}$  is

$$\vec{AB} = \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \\ z_2 - z_1 \end{pmatrix} = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$

### Modulus of a vector

The modulus of a vector is the length of the directed line segment represented by the vector.

If  $P(x, y)$  is a point in the plane then modulus of vector  $OP$  is

$$|\vec{OP}| = OP = \sqrt{x^2 + y^2}$$

Similarly, If  $P(x, y, z)$  is a point in the space then modulus of vector  $OP$  is

$$|\vec{OP}| = OP = \sqrt{x^2 + y^2 + z^2}$$

If  $A(x_1, y_1)$  and  $B(x_2, y_2)$  are two points in a plane then modulus of vector  $AB$  is

$$|\vec{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Similarly, If  $A(x_1, y_1, z_1)$  and  $B(x_2, y_2, z_2)$  are two points in the space then modulus of vector  $AB$  is

$$|\vec{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

### Different types of vectors

|                         |                                                                                                                          |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Unit vector</b>      | A vector with magnitude 1 (unity) is known as a unit vector.                                                             |
| <b>Null/zero vector</b> | A vector whose magnitude is zero, is known as zero vector.                                                               |
| <b>Negative vector</b>  | A negative vector is simply a vector that has the same magnitude as a given vector but points in the opposite direction. |
| <b>Like vectors</b>     | Two vectors are said to be like vectors having the same direction.                                                       |
| <b>Unlike vectors</b>   | Two vectors are said to be unlike vectors having the opposite direction.                                                 |
| <b>Equal vectors</b>    | Two vectors are said to be equal if their magnitudes are equal and directions are same.                                  |

### Addition of Two Vectors

If  $\vec{a} = (a_1, a_2)$  and  $\vec{b} = (b_1, b_2)$  then

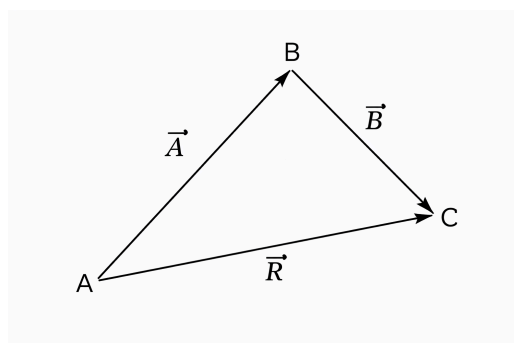
$$\vec{a} + \vec{b} = (a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 + b_2)$$

### Triangle Law of Vector Addition

If  $\vec{AB} = \vec{A}$ ,  $\vec{BC} = \vec{B}$  and  $\vec{AC} = \vec{R}$  then

$$\vec{AB} + \vec{BC} = \vec{AC}$$

$$\Rightarrow \vec{A} + \vec{B} = \vec{R}$$



### Difference of Two Vectors

If  $\vec{a} = (a_1, a_2)$  and  $\vec{b} = (b_1, b_2)$  then

$$\vec{a} - \vec{b} = (a_1, a_2) - (b_1, b_2) = (a_1 - b_1, a_2 - b_2)$$

### Multiplication of a vector by a scalar

If  $\vec{a} = (a_1, a_2)$  is a vector and  $k$  is a scalar then their product is defined by

$$k\vec{a} = k(a_1, a_2) = (ka_1, ka_2)$$

### Laws of vector algebra

If  $\vec{a}, \vec{b}, \vec{c}$  are three vectors and  $m$  is any scalar then we have the following laws:

#### 1. Commutative law

$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$

#### 2. Associative law

$$\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$$

#### 3. Distributive law

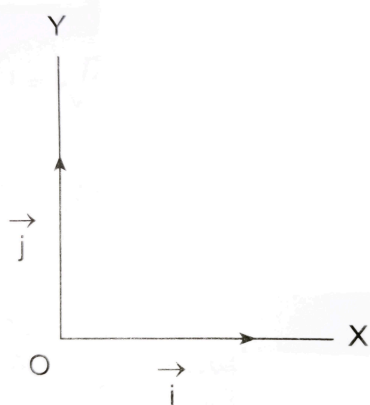
$$m(\vec{a} + \vec{b}) = m\vec{a} + m\vec{b}$$

### The unit vectors $\vec{i}$ , $\vec{j}$ , and $\vec{k}$

If  $\vec{i}$  and  $\vec{j}$  are two unit vectors along x-axis and y-axis respectively then

$$\vec{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\vec{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

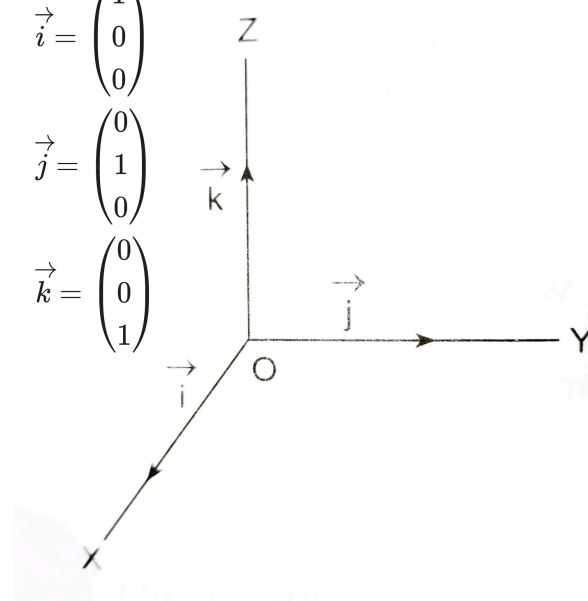


If  $\vec{i}$ ,  $\vec{j}$  and  $\vec{k}$  are three unit vectors along x-axis, y-axis and z-axis respectively then

$$\vec{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\vec{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\vec{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$



### Unit vector along given vector

If  $\vec{a} = a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k}$  is vector then the unit vector along its direction is

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k}}{\sqrt{a_1^2 + a_2^2 + a_3^2}}$$

Corollary: The direction cosines of the line represented by vector  $a$  are

$$\frac{a_1}{\sqrt{a_1^2 + a_2^2 + a_3^2}}, \frac{a_2}{\sqrt{a_1^2 + a_2^2 + a_3^2}}, \frac{a_3}{\sqrt{a_1^2 + a_2^2 + a_3^2}}$$

### #Collinear and Non-collinear Vectors

Collinear vectors are the vectors that lie on the same straight line or are parallel to the same line.

All collinear vectors are parallel, but all parallel vectors are not collinear. Non-collinear vectors are the vectors that do not lie on the same straight line and are not parallel to the same line.

### Condition of Collinearity

If a vector  $\vec{a}$  is collinear with a given vector  $\vec{b}$  then  $\vec{a} = k \vec{b}$  for some scalar  $k$ .

### #Coplanar and Non-coplanar Vectors

Coplanar vectors are the vectors that lie on the same plane or are parallel to the same plane.

Non-coplanar vectors are the vectors that do not lie on the same plane and are not parallel to the same plane.

### Condition of Coplanarity

If  $\vec{a}, \vec{b}, \vec{c}$  are three coplanar vectors then  $\vec{a} = m \vec{b} + n \vec{c}$  for some scalars  $m$  and  $n$ .

### #Linear Combination of Vectors

A linear combination of vectors is a new vector formed by scaling a set of vectors by scalars and adding them.

If a vector  $\vec{v}$  can be expressed as  $\vec{v} = c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_n \vec{v}_n$  for some scalars  $c_1, c_2, \dots, c_n$  then the vector  $\vec{v}$  is said to be the linear combination of vectors  $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ .

If  $\vec{v} = a \vec{i} + b \vec{j} + c \vec{k}$  then  $\vec{v}$  is said to be the linear combination of  $\vec{i}, \vec{j}, \vec{k}$ .

### #Linearly Dependent and Independent Vectors

| Linearly Dependent Vectors                                                                                         | Linearly Independent Vectors                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| A set of vectors is said to be linearly dependent if at least one vector is a linear combination of other vectors. | A set of vectors is said to be linearly independent vectors if no vector in the set can be written as a linear combination of other vectors. |

### Condition for Linear Dependency and Independency

Two vectors  $\vec{a} = (a_1, a_2)$  and  $\vec{b} = (b_1, b_2)$  are said to be linearly dependent vectors if  $\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = 0$  else they are linearly independent vectors.

Three vectors  $\vec{a} = (a_1, a_2, a_3)$ ,  $\vec{b} = (b_1, b_2, b_3)$  and  $\vec{c} = (c_1, c_2, c_3)$  are said to be linearly dependent

vectors if  $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$  else they are linearly independent vectors.